Compositional system

geometry and dynamics for,

IIASA NGFS Phase-4 NZ-2050 sectors

2020-2050

Report title	Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sector
Generated	2026-05-06
Run ID	v1.1.x-final
Dataset ID	iiasa_ngfs
T (records)	31
D (carriers)	7
Engine version	cnt 2.0.4
Schema version	2.1.0
Engine signature (sha256[:32])	97748868a0381cb7b995d2ae562e5c42…
Source CSV SHA-256	c22adce0bfa1aad3991cea2e9849cea0…
Closed data SHA-256	acba22c6cfda1f0a3ba016fb08ba55d5…
Content SHA-256 (canonical JSON)	4991f57d5588d1dc12748d37b922121c…

```
Reading order: geometry first, dynamics second.
  §A  Data disclosure & processing notes
  §B  Geometry : structure on the simplex
              evolution of proportions → ternaries → variation
              → correlation → scree → biplot → top-N biplot
              → balance-dendrogram → SBP table
  §C  Dynamics : evolution through compositional space
              trajectories (Hs, ‖clr‖, ILR axes) → System Course Plot
              → bearing rose → bearing heatmap → ω → metric trace
              → helmsman → ring class → locks → CBS
              → pairwise divergence ranking → ω↔Hs
  §D  Summary
```

Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 2 | §A Data Disclosure & Processing-Anomaly Review

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

```
INPUT DATA
  source file        : iiasa_ngfs_input.csv
  source SHA-256     : c22adce0bfa1aad3991cea2e9849cea0a0a21f06acb794837acbdcd74750d4b8
  closed data SHA-256: acba22c6cfda1f0a3ba016fb08ba55d5075be0d6fa87994df3e1c6cc9bc885d4
  records (T)        : 31
  carriers (D)       : 7
  carriers           : Energy, Transport, Industry, Buildings, Agriculture, LULUCF, Other

ORDERING
  is_temporal        : True
  ordering_method    : by-time

ZERO REPLACEMENT
  zeros replaced     : 0
  replacement δ      : 1e-15
  method             : multiplicative

UNITS LINEAGE  (schema 2.1.0)
  input_units                 : ratio
  higgins_scale_units         : neper
  units_scale_factor_to_neper : 1.0

ENGINE CONFIG ACTIVE OVERRIDES
  (none — defaults from cnt.py USER CONFIGURATION block)
```

```
ENGINE SOURCE FINGERPRINT
  stage2_locked.py      : c318ab82c15890b947c3fc15a6c66b55aa23f3cf94104cf9b9d5fbdb
  cnt.py                : 1e10cb62d6588ac9557c6a2e8436c2980f0c35b379f29c05c0377711
  combined_sha256       : 97748868a0381cb7b995d2ae562e5c429988ee7fd556943b882c5fcb

DEGENERACY FLAGS
  LCF_flag : {'flag': 'pre_aligned_compositionally', 'severity': 'warning', 'message': 'Rec

LOCK EVENTS  (period-2 detector)
  total events       : 0

EITT BENCH-TEST RESIDUALS
  H_mean_full           : 1.7609003750938372
  M_sweep               : [{'M': 2, 'n_blocks': 16, 'H_mean_decimated': 1.76114199
  gate_pct              : 5.0
  note                  : Empirical observation of trajectory smoothness under tem

SANITY CHECKS  (computed during this report)
  ‖clr‖ non-finite count : 0
  ω    non-finite count : 0
  Hs   non-finite count : 0
  Σ CLR per row range   : [-2.89e-15, +3.55e-15]   (should be ≈ 0)
```

If any flag above is non-empty, the analysis on subsequent pages still ran but the reader should weight those plates accordingly. Determinism is preserved either way.

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Plate 3 | Evolution of Proportions (raw + cumulative)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

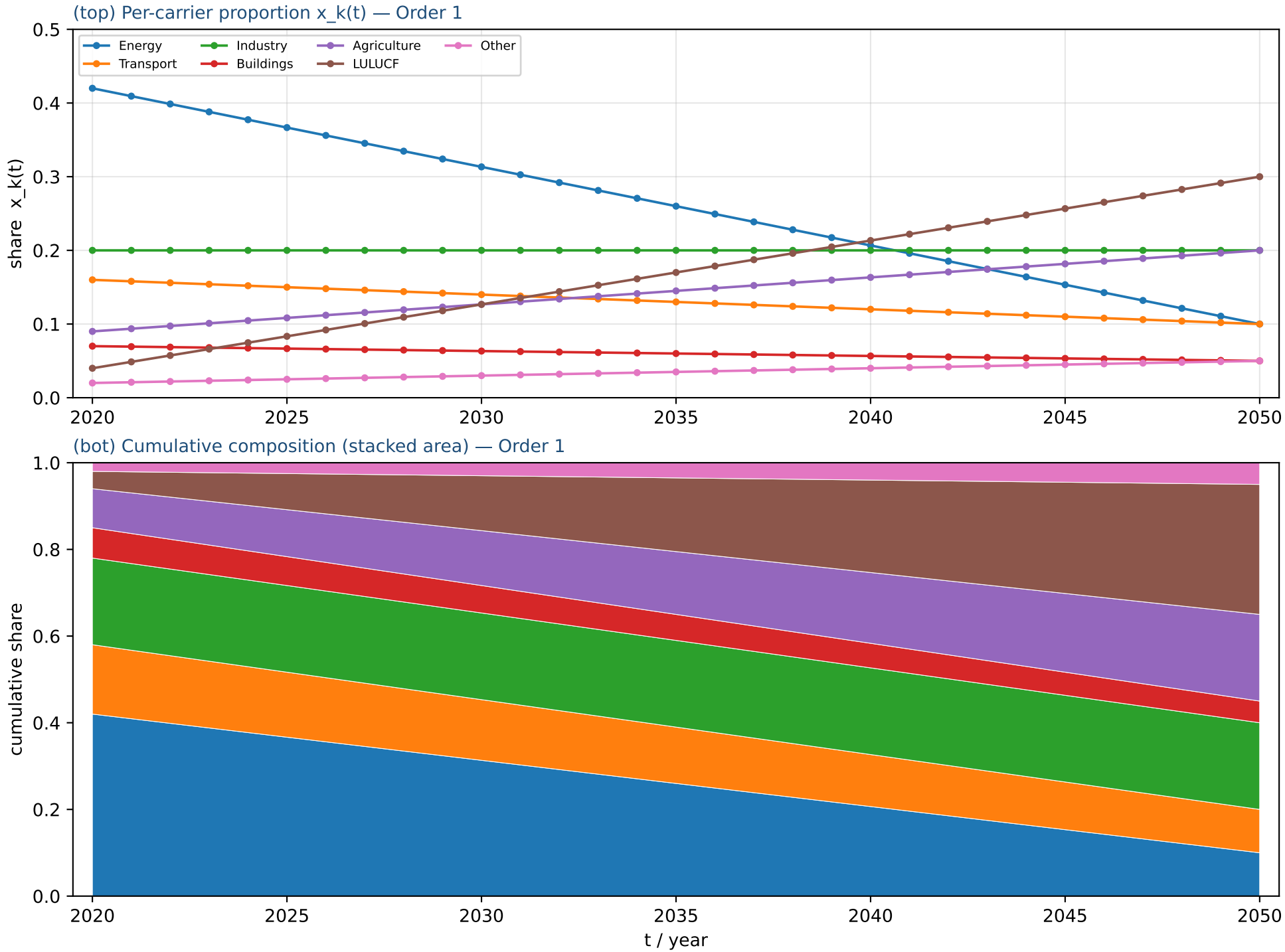

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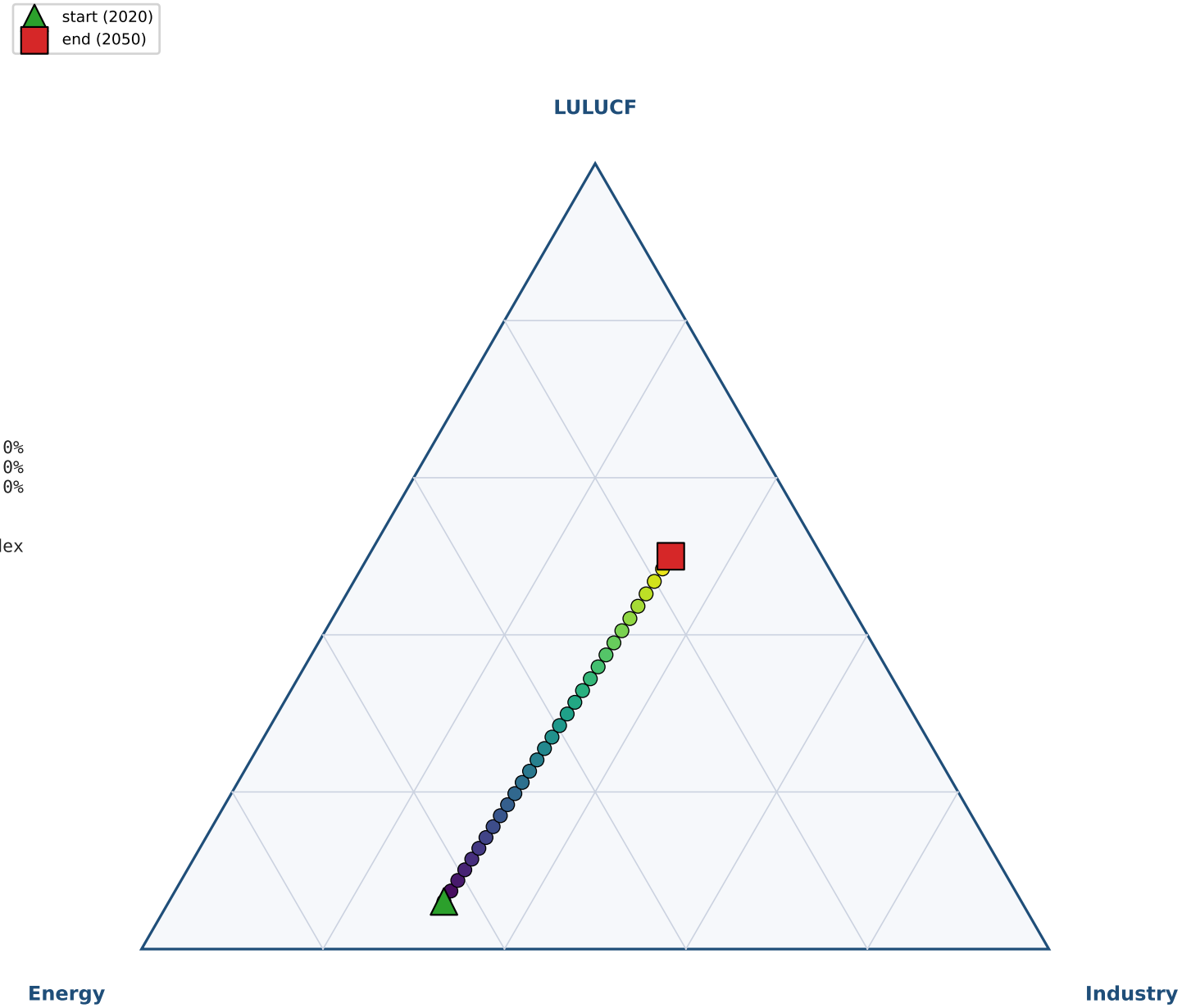
Plate 4 | Ternary Diagram (top-3 carriers)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

3-part subcomposition closed over Energy, Industry, LULUCF
(remaining 4 carriers absorbed into closure)




```
Top-3 by mean share:
  Energy          26.0%
  Industry        20.0%
  LULUCF          17.0%

T = 31 timesteps
colour map = time index
▲ start  ■ end
```

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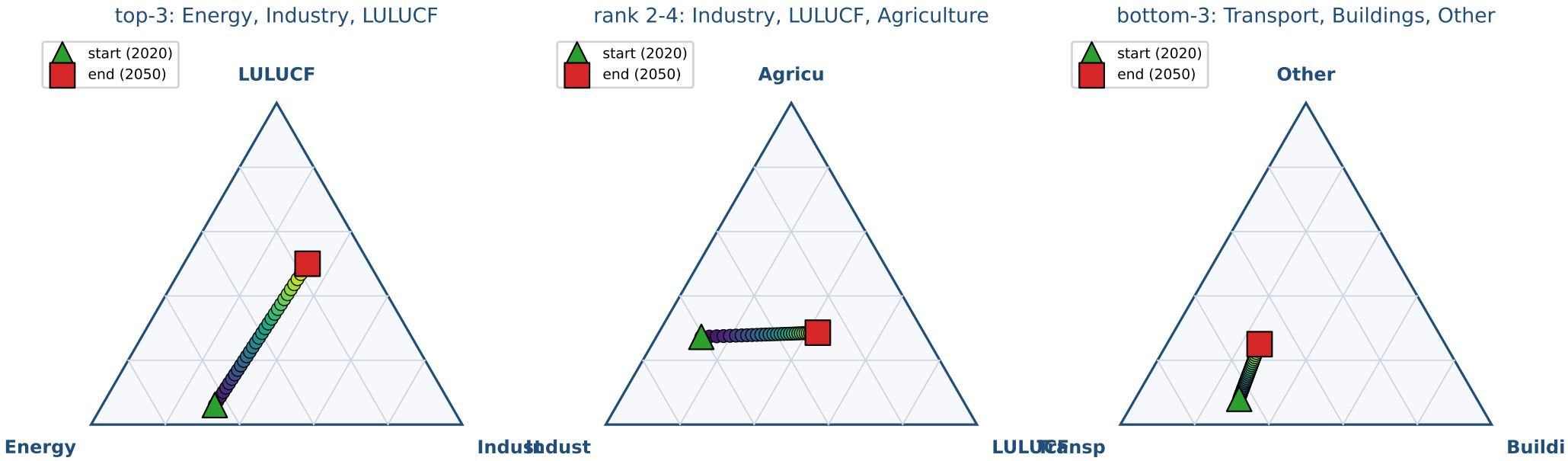
Plate 5 | Three-Part Subcompositions (3 ternaries)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Each ternary closes the chosen 3 carriers; remaining carriers absorbed into closure. ▲ start ■ end | colour map = time index

Triplet source: default heuristic

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Plate 6 | Variation matrix τ_ij heatmap

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

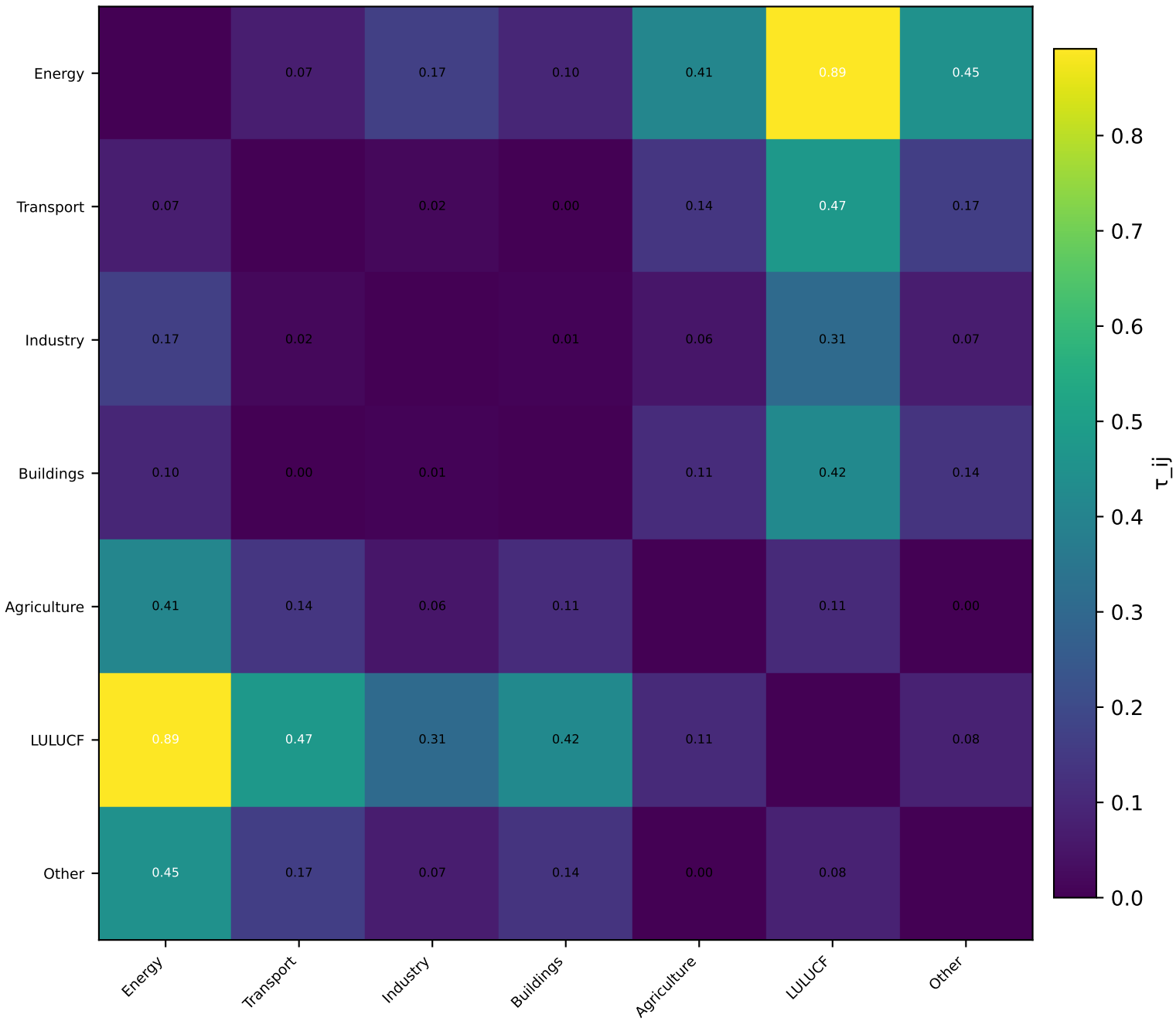

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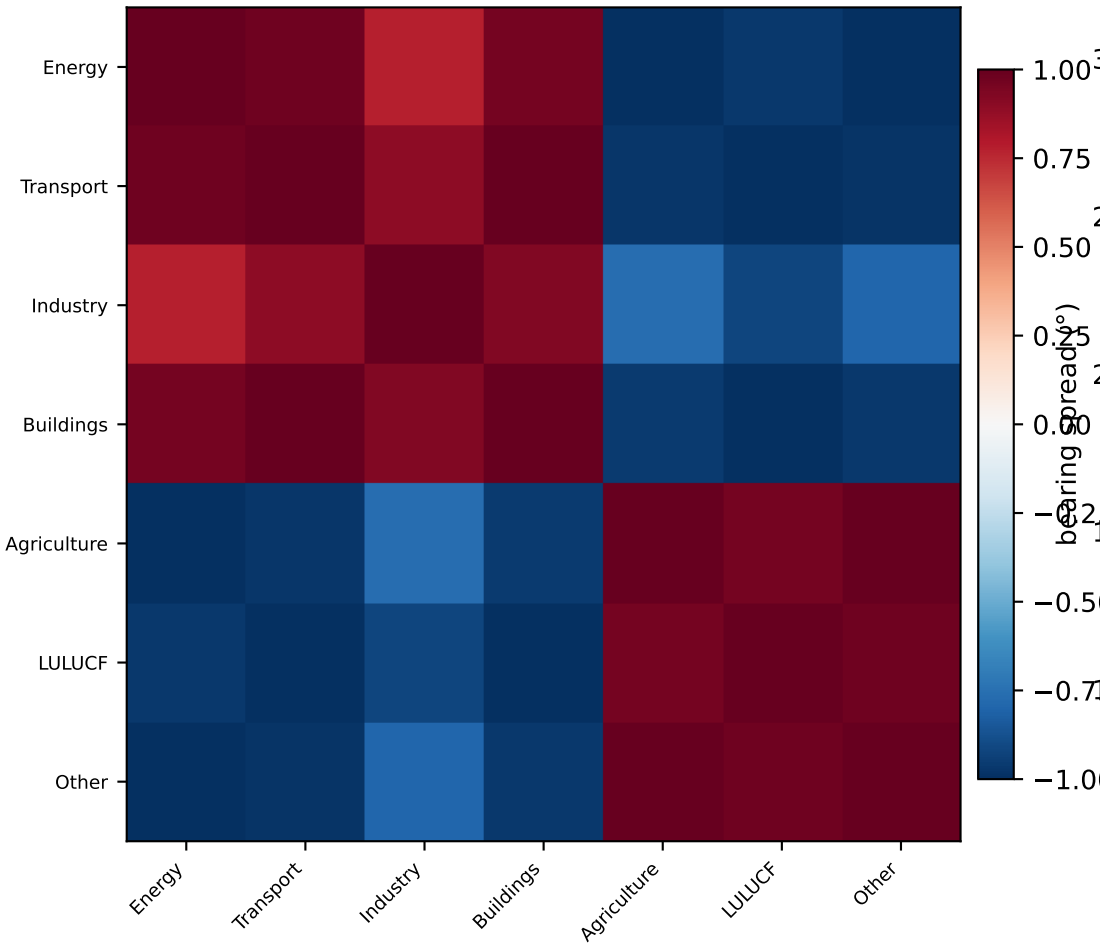
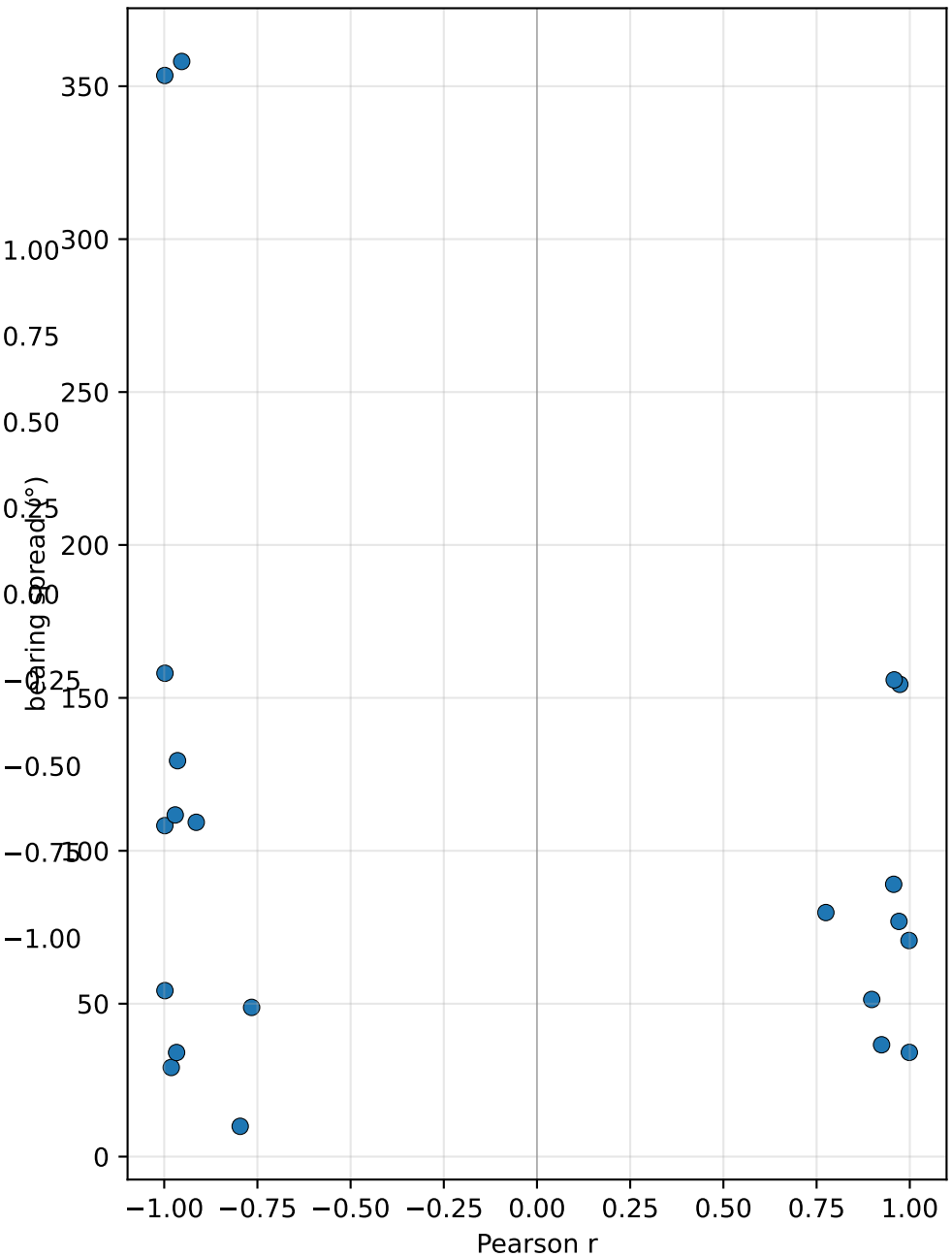
Plate 7 | Pair Pearson r heatmap + co-movement scatter

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

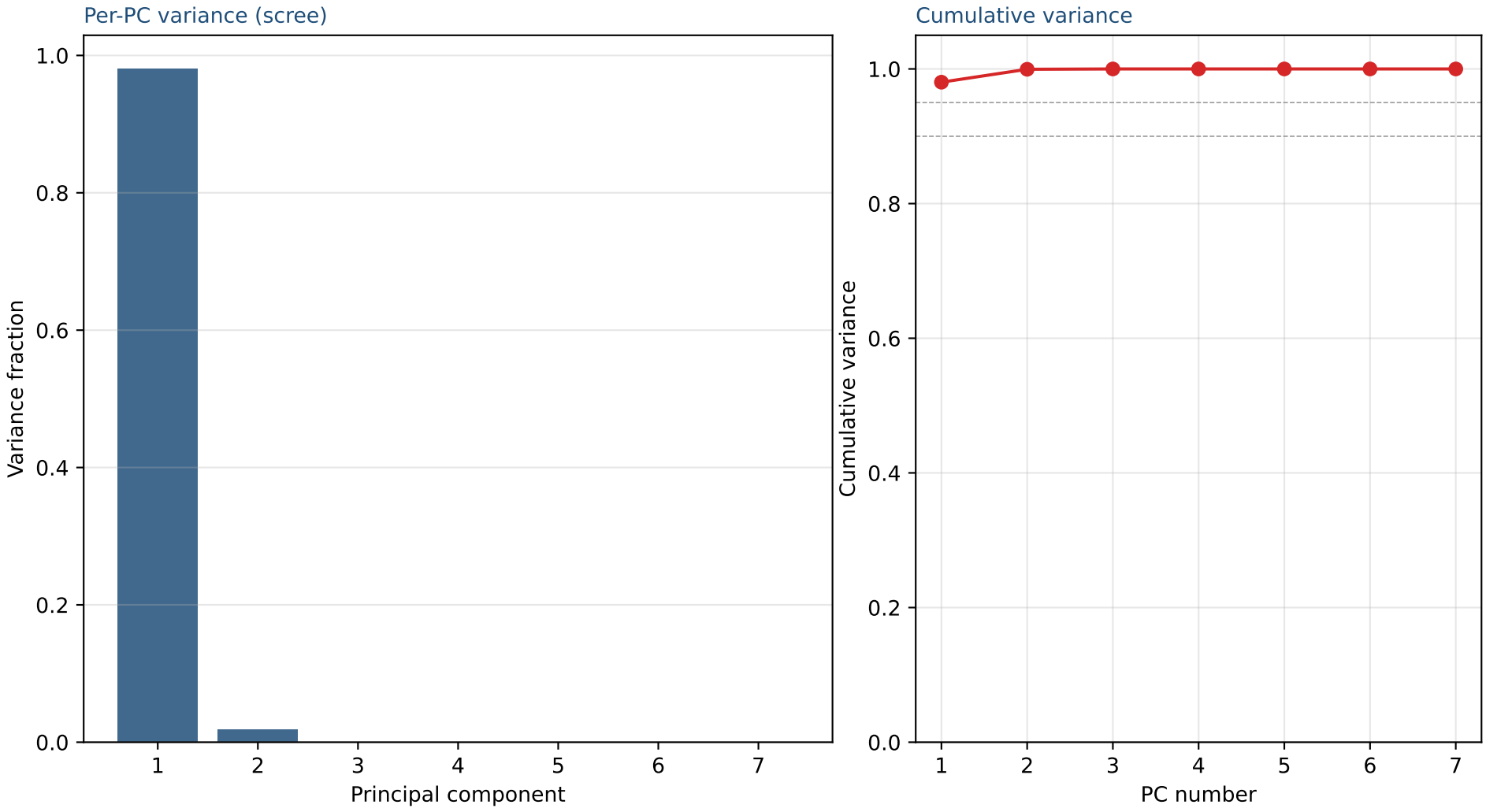




Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 8 | Scree Plot - CLR PCA

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



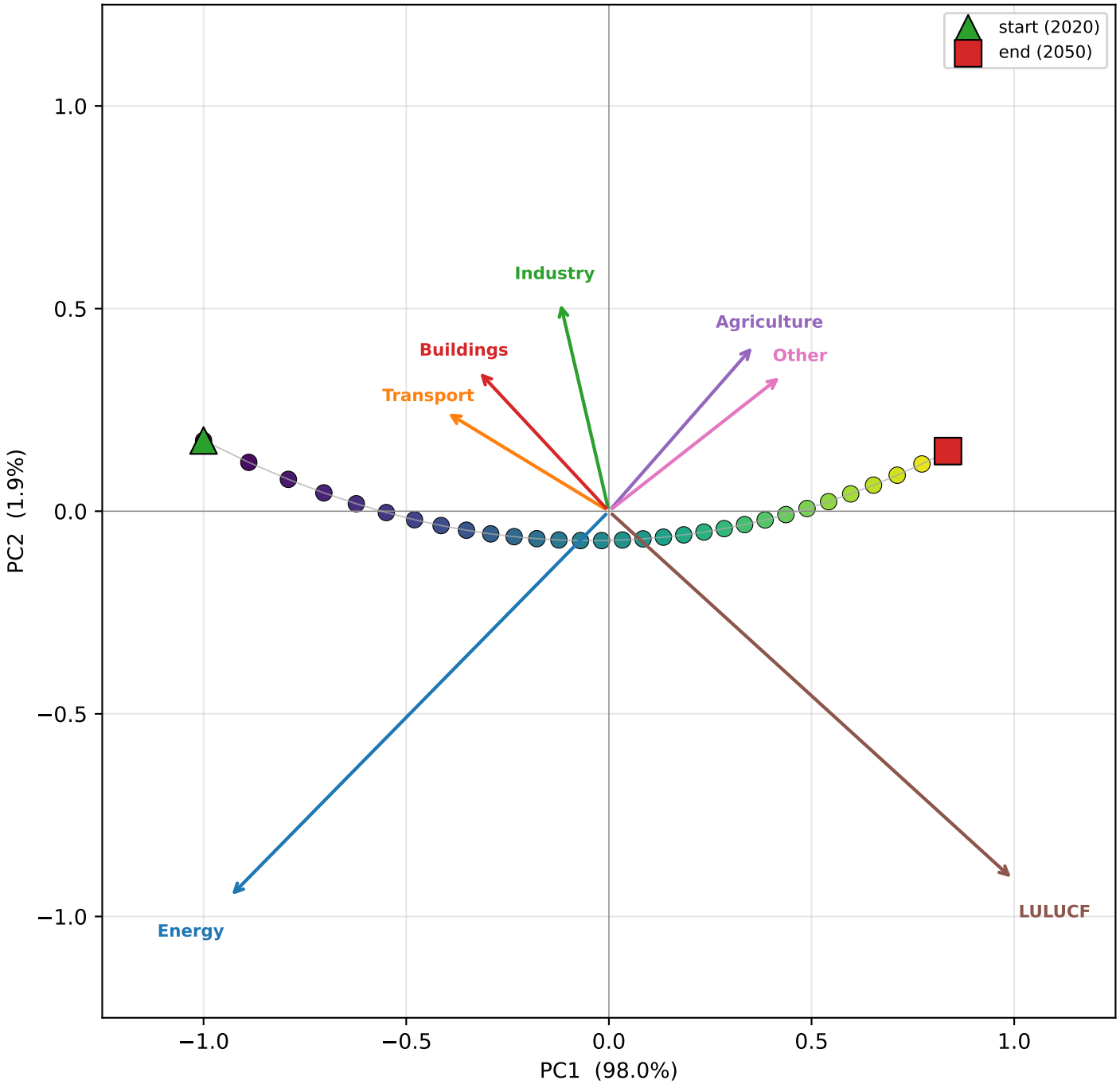

```
Per-PC variance contribution:
  PC1: 98.03%   cum 98.03%
  PC2:  1.92%   cum 99.95%
  PC3:  0.05%   cum 100.00%
  PC4:  0.00%   cum 100.00%
  PC5:  0.00%   cum 100.00%
  PC6:  0.00%   cum 100.00%
  ... 1 more PCs
```

Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 9 | CLR Biplot (CoDa biplot)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

CoDa biplot - covariance form, 2D variance explained 100.0%



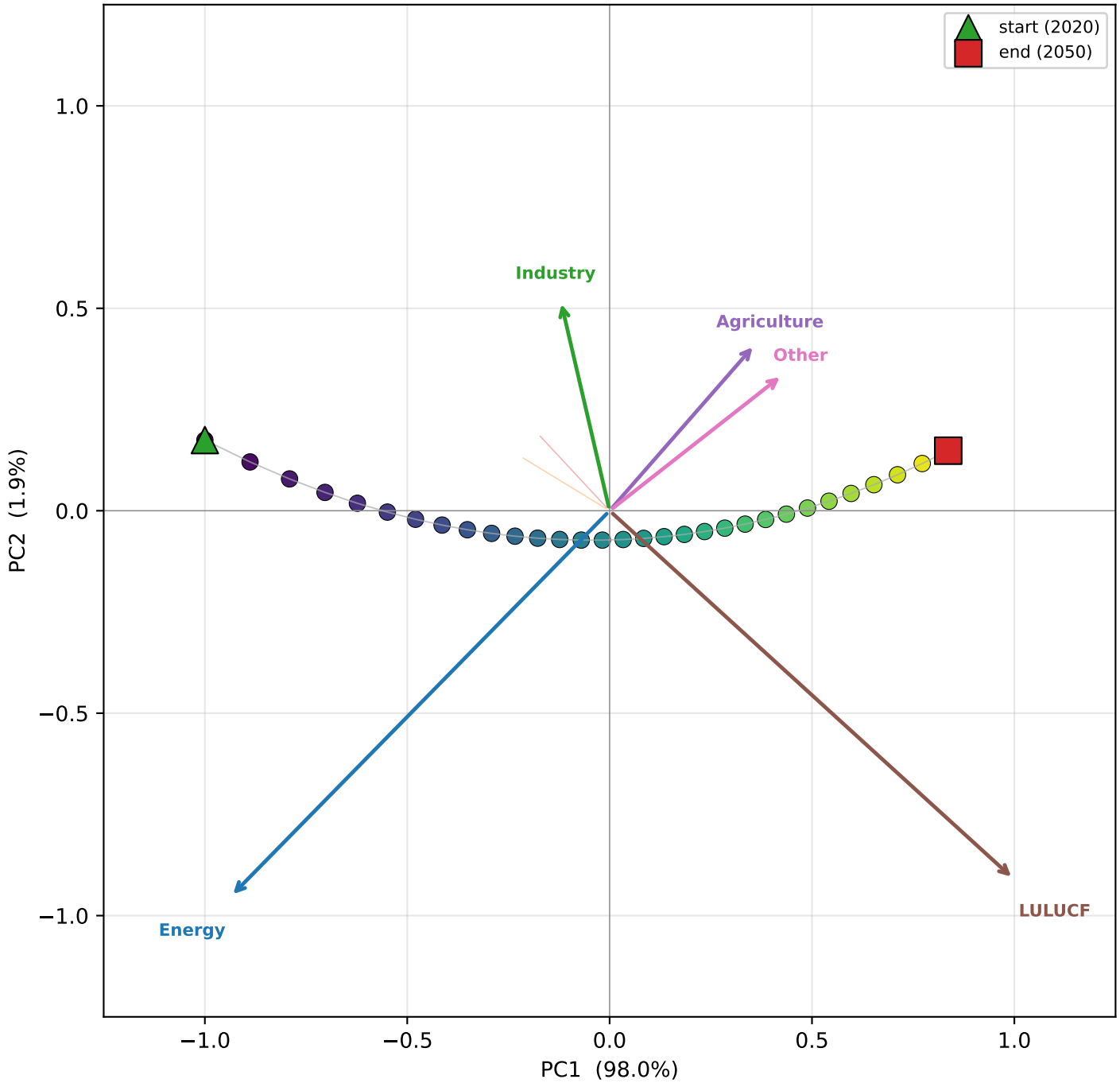

```
CoDa biplot (Aitchison-Greenacre):
  - Points: timestep scores in PC(1,2) of c
  - Rays:   carrier loadings (columns of V)
  - Ray length = carrier importance on the
  - Angle between rays ~ negative log-ratio
  - Distance between point projections = ap
    of the Aitchison distance restricted to

PC1 captures 98.0% of variance.
PC2 captures 1.9% of variance.
Total 2D plane:  100.0%
```

Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 10 | CLR Biplot — top-5 carriers by loading magnitude

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

Top-5 biplot — 2D variance 100.0% (D = 7, dropped 2 short rays)



Loading magnitudes $\sqrt{(G_1^2 + G_2^2)}$:

+ LULUCF	0.8594
+ Energy	0.8500
+ Agriculture	0.3477
+ Other	0.3463
+ Industry	0.3397

Dropped (faint stubs only):

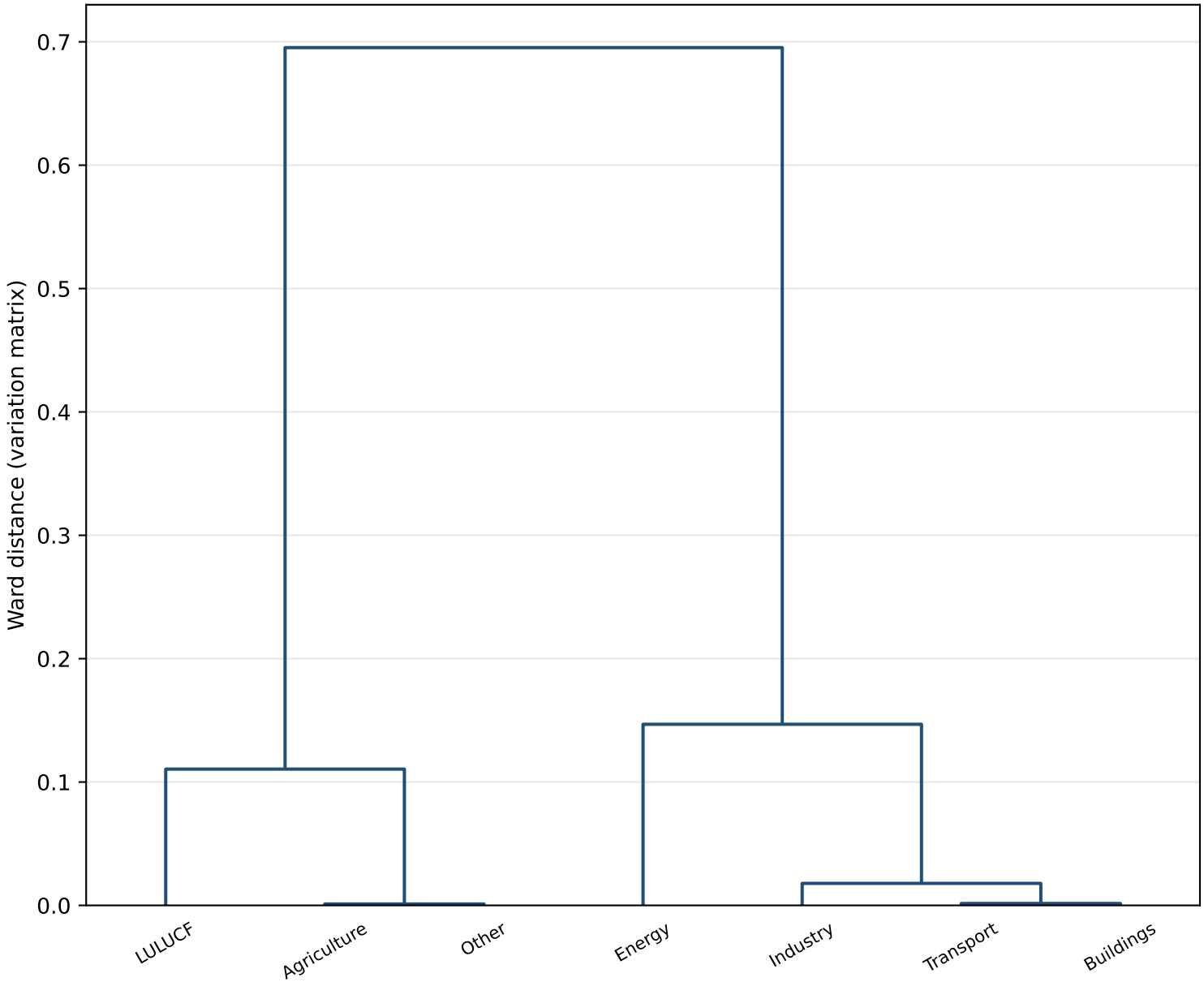
Buildings	0.3031
Transport	0.3013

Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 11 | Balance-Dendrogram (Ward on variation matrix)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

Hierarchical balance tree — each merge defines one balance partition




```
Balance-dendrogram (Pawlowsky-Glahn-
  – Ward clustering on the (symmetri
  – Each internal node merges two ca
  – That merge defines a balance:
      b = sqrt(|A||B|/(|A|+|B|)) ·
    where g() is the geometric mean.
  – The balances form an orthonormal

D = 7 carriers → D-1 = 6 balances.
Tightly-clustered carriers (low vari
See companion plate for the full SBP
```

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Plate 12 | Sequential Binary Partition (SBP) Matrix

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

SBP cells: + = numerator, - = denominator, . = not in this balance

	b1	b2	b3	b4	b5	b6
Energy	.	.	.	.	+	-
nsport	.	+	-	.	-	-
dustry	.	.	+	.	-	-
ldings	.	-	-	.	-	-
ulture	+	.	.	-	.	+
LULUCF	.	.	.	+	.	+
Other	-	.	.	-	.	+

Balance b_k = sqrt(|A_k||B_k|/(|A_k|+|B_k|)) * ln(g(A_k) / g(B_k)) | D = 7; D-1 = 6 balances.

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Plate 13 | Hs (Higgins scale) trajectory — Order 2 aggregate

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

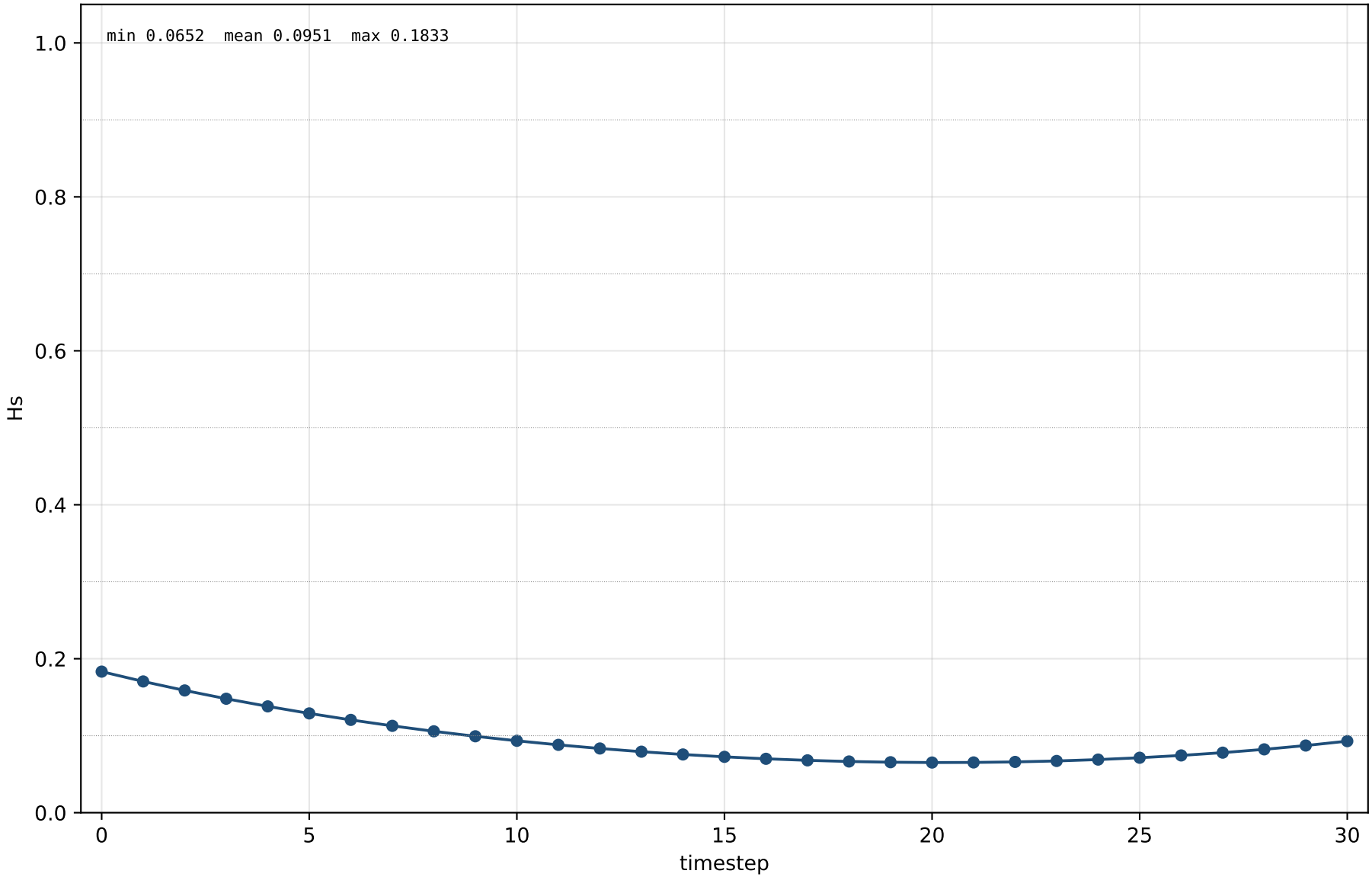

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Plate 14 | Aitchison norm trajectory ‖clr(x)‖₂ — Order 2 aggregate

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

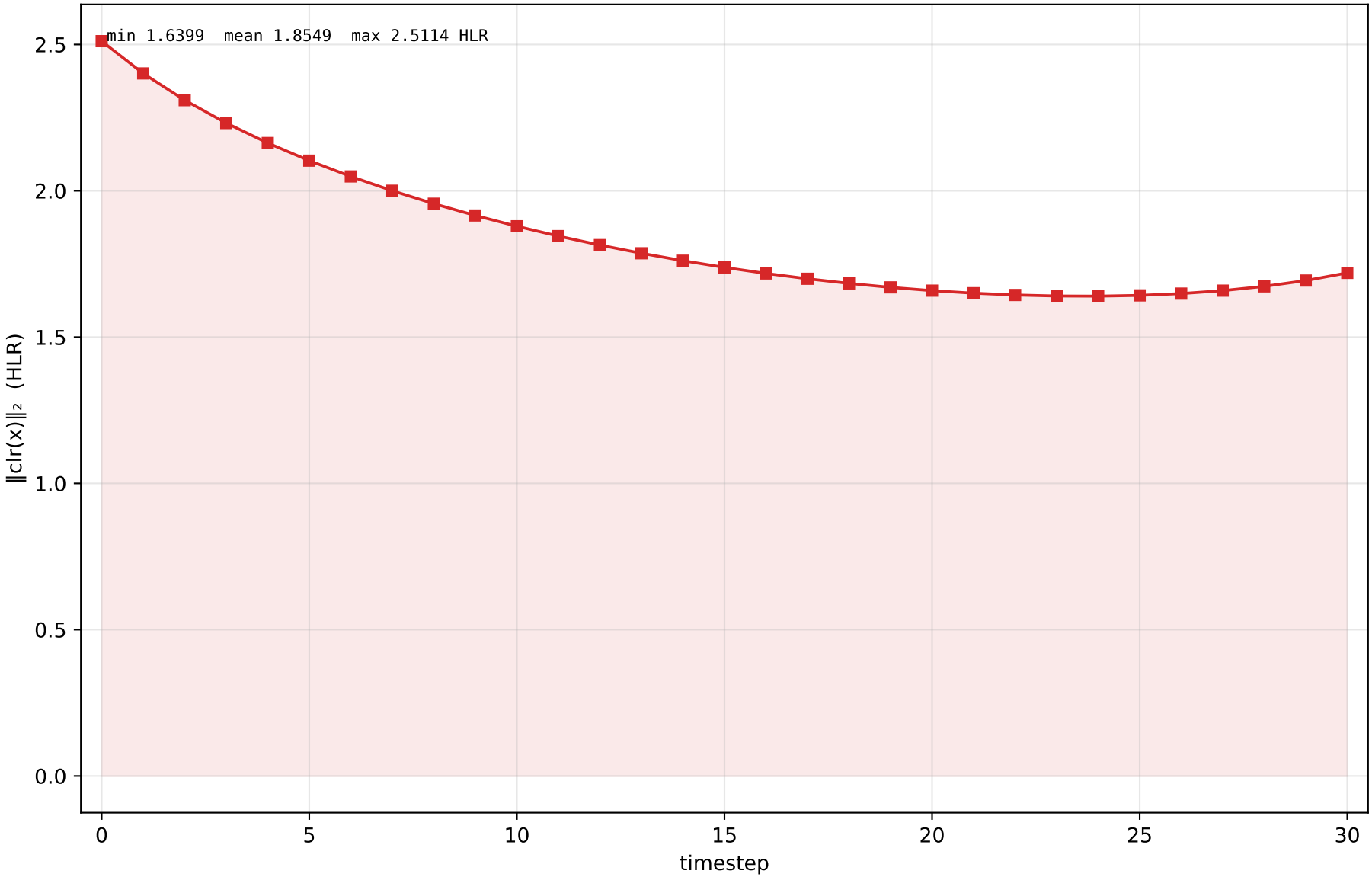

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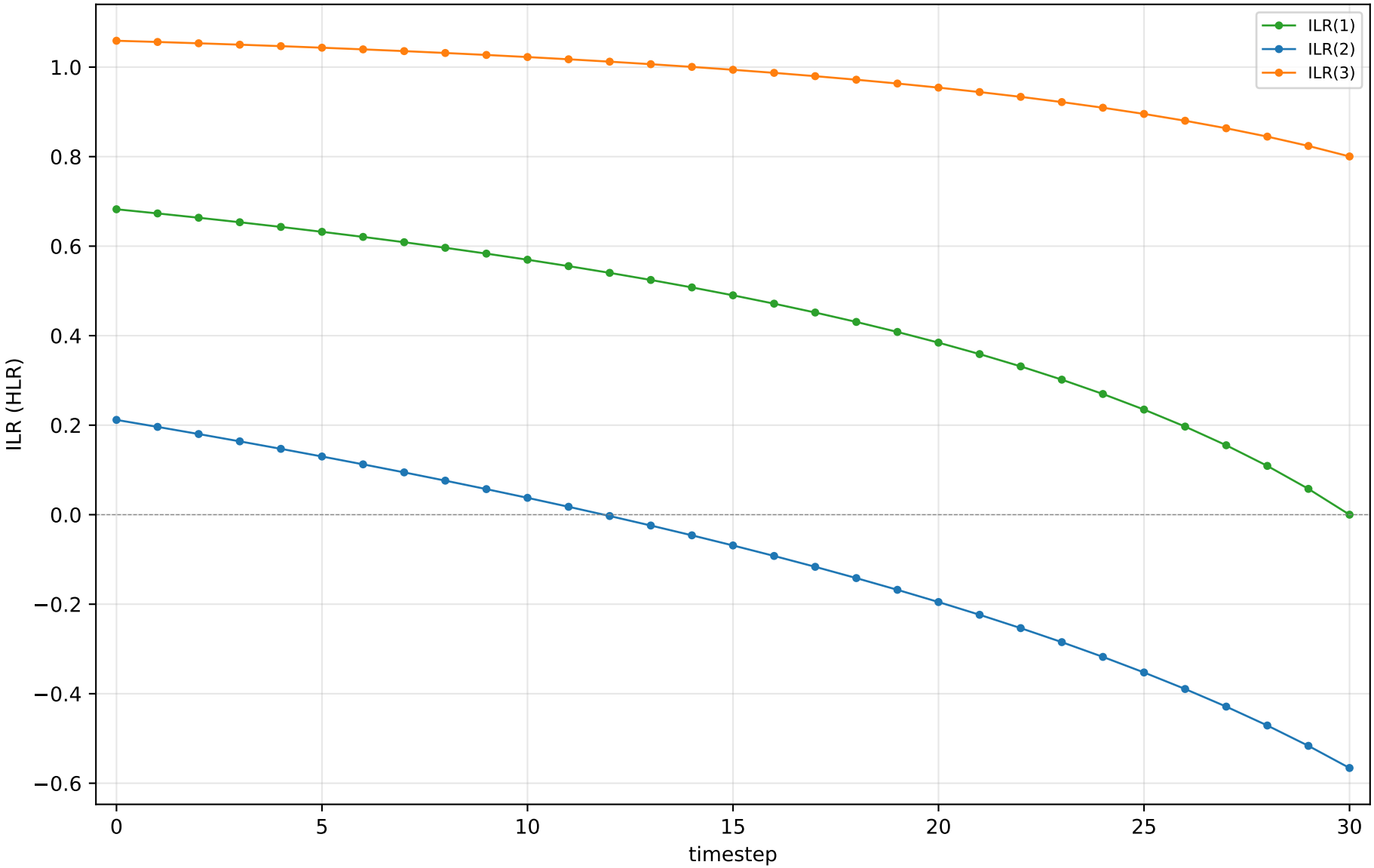
Plate 15 | ILR axes 1–3 trajectories (Helmert basis) — Order 2 aggregate

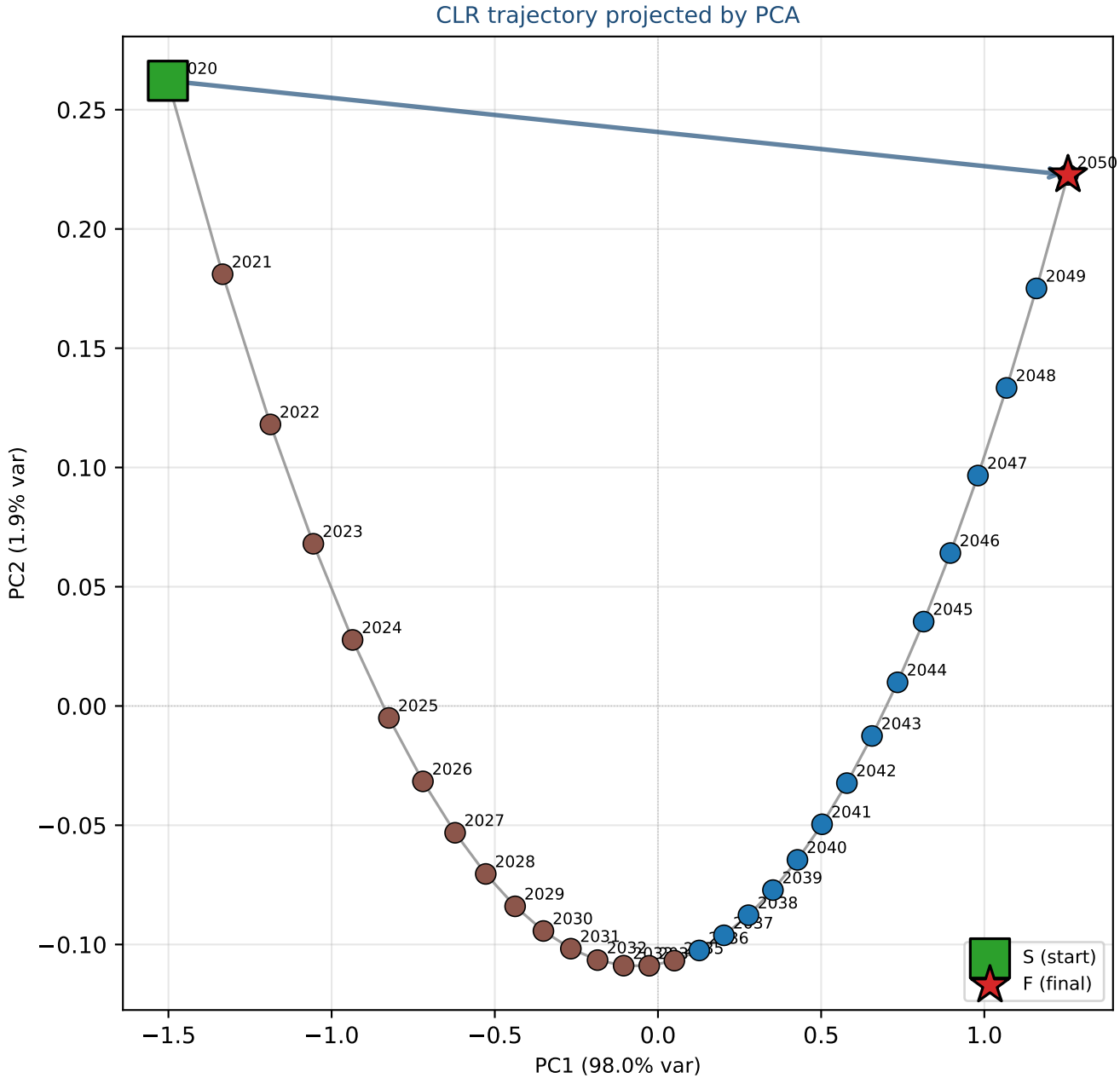
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 16 | System Course Plot — PCA 2D path with S→F net vector

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



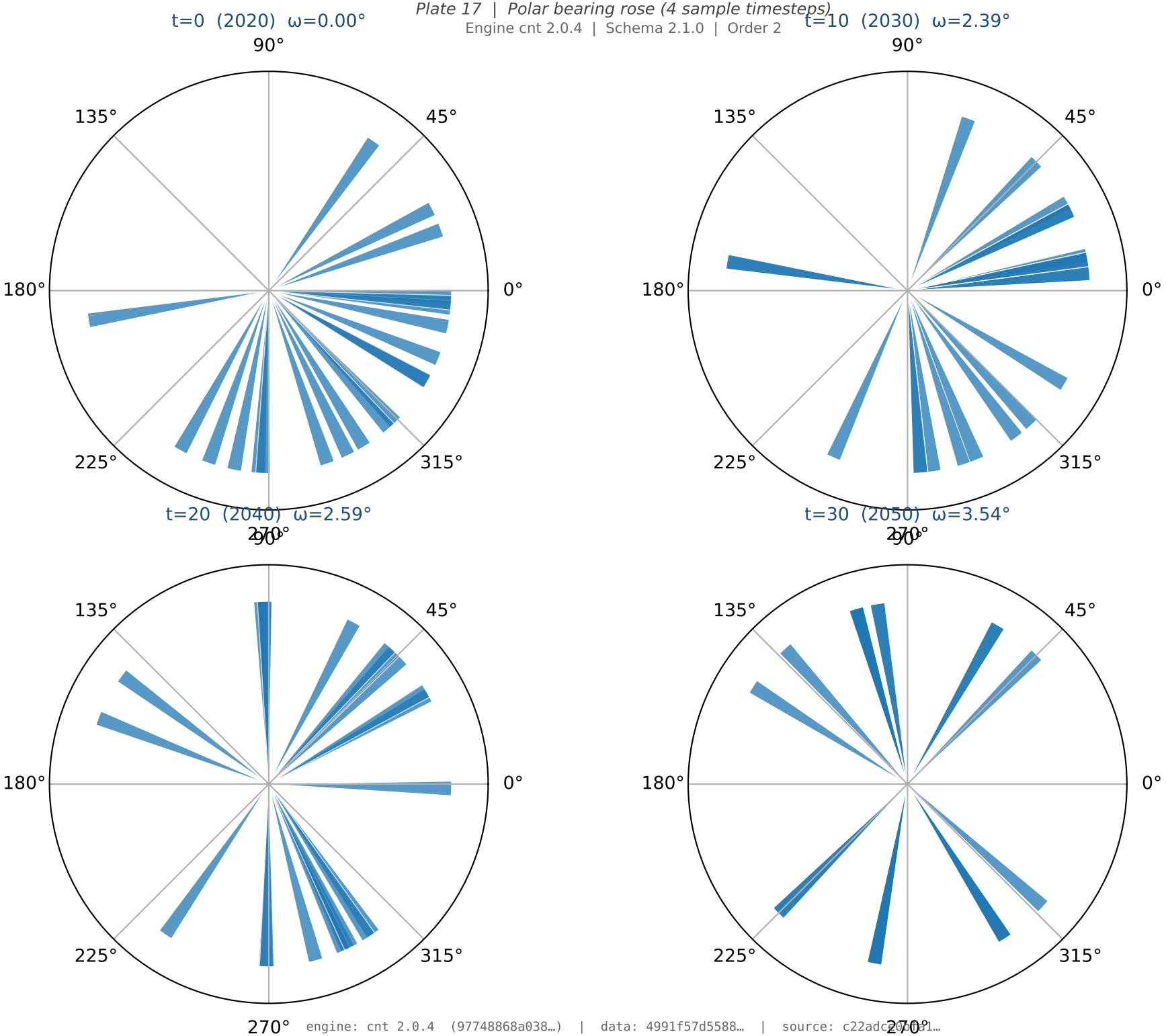
Navigation metrics (V_net = h_F − h_S)

```
h_start label    : 2020
h_final label    : 2050

net_distance     : 2.7594 HLR
path_length      : 2.8803 HLR
course_directness: 0.9580
  (1.0 = straight line, 0 = pure looping)

dynamic range S  : 3.0445 HLR
dynamic range F  : 1.7918 HLR

Reading guide:
  tight cluster  → stable composition
  long excursion → structural shift
  sharp turn     → regime transition
  loop           → cyclic behaviour
  spiral         → systematic drift
```

Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 17 | Polar bearing rose (4 sample timesteps)

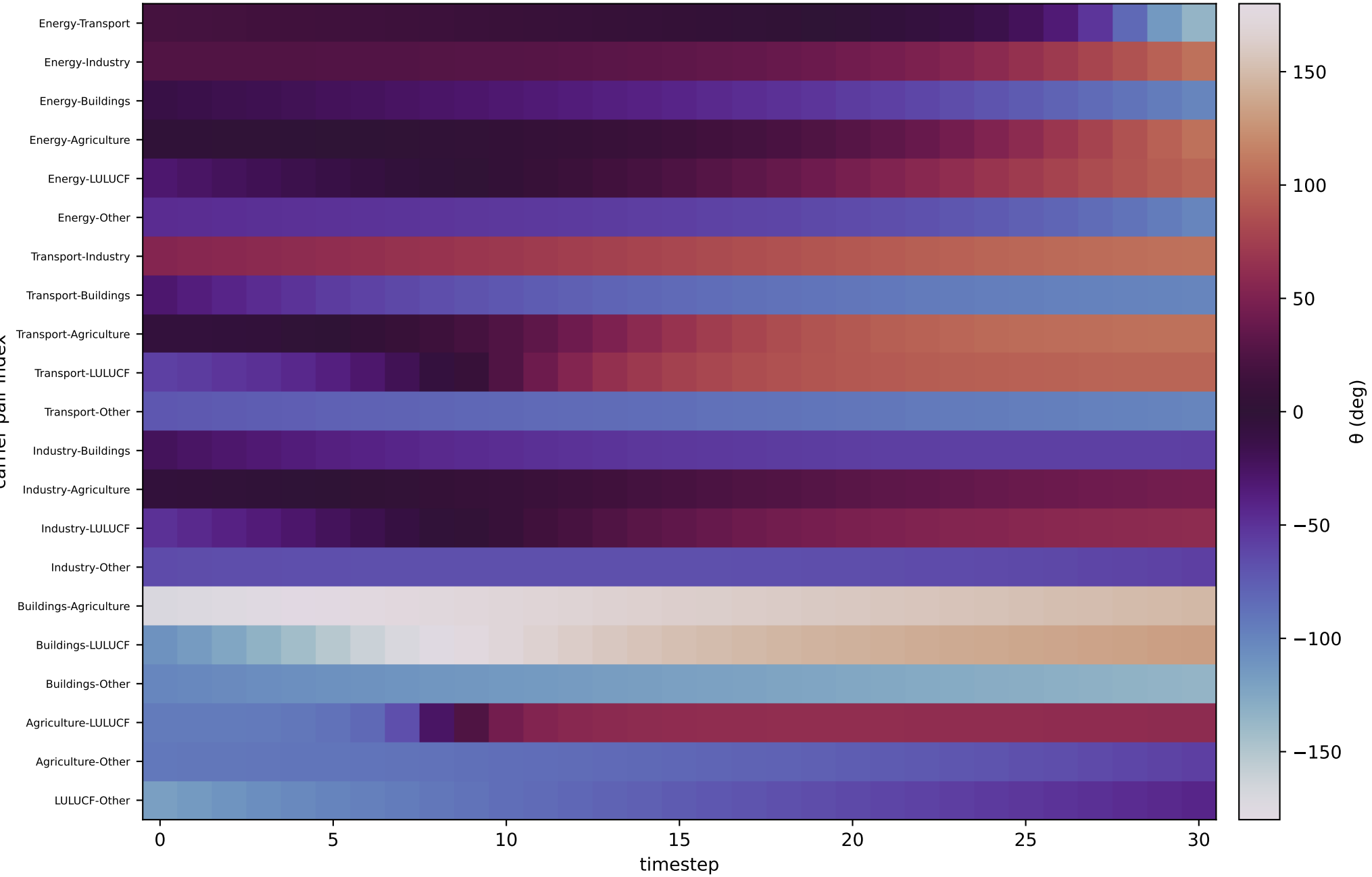
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 18 | Bearing heatmap T × C(D,2) pairs

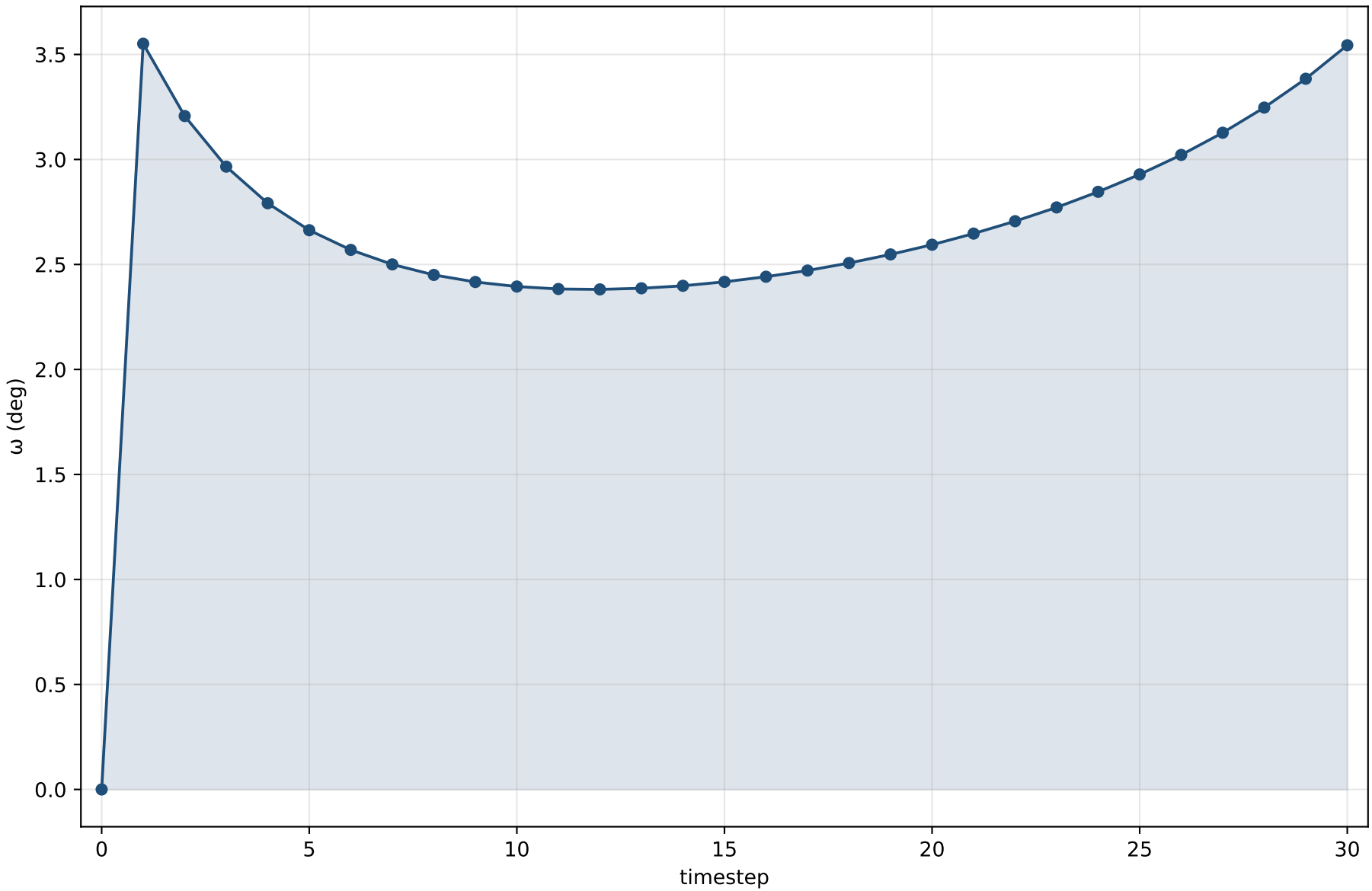
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 19 | ω angular velocity trajectory

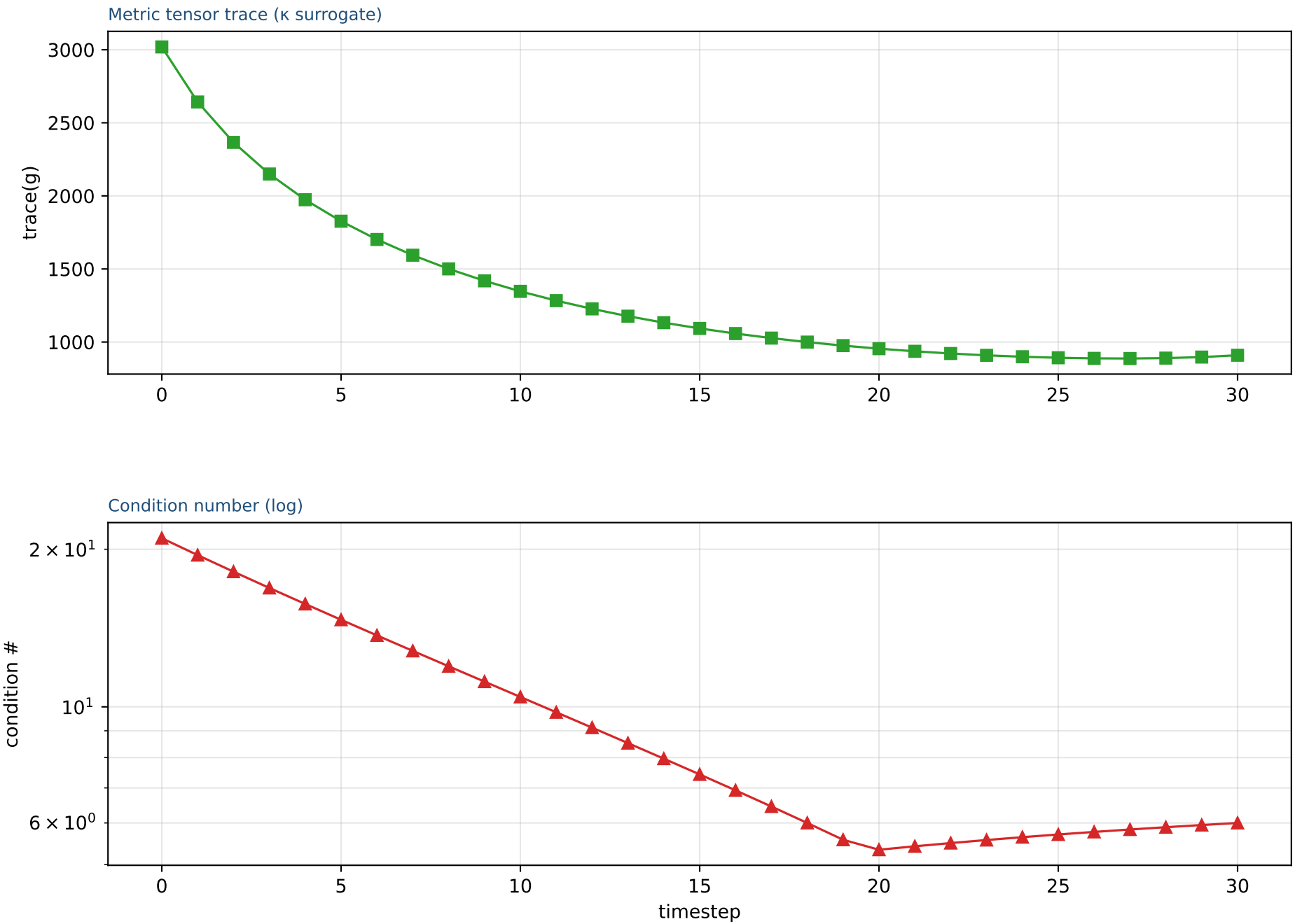
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 20 | Metric tensor trace + condition number

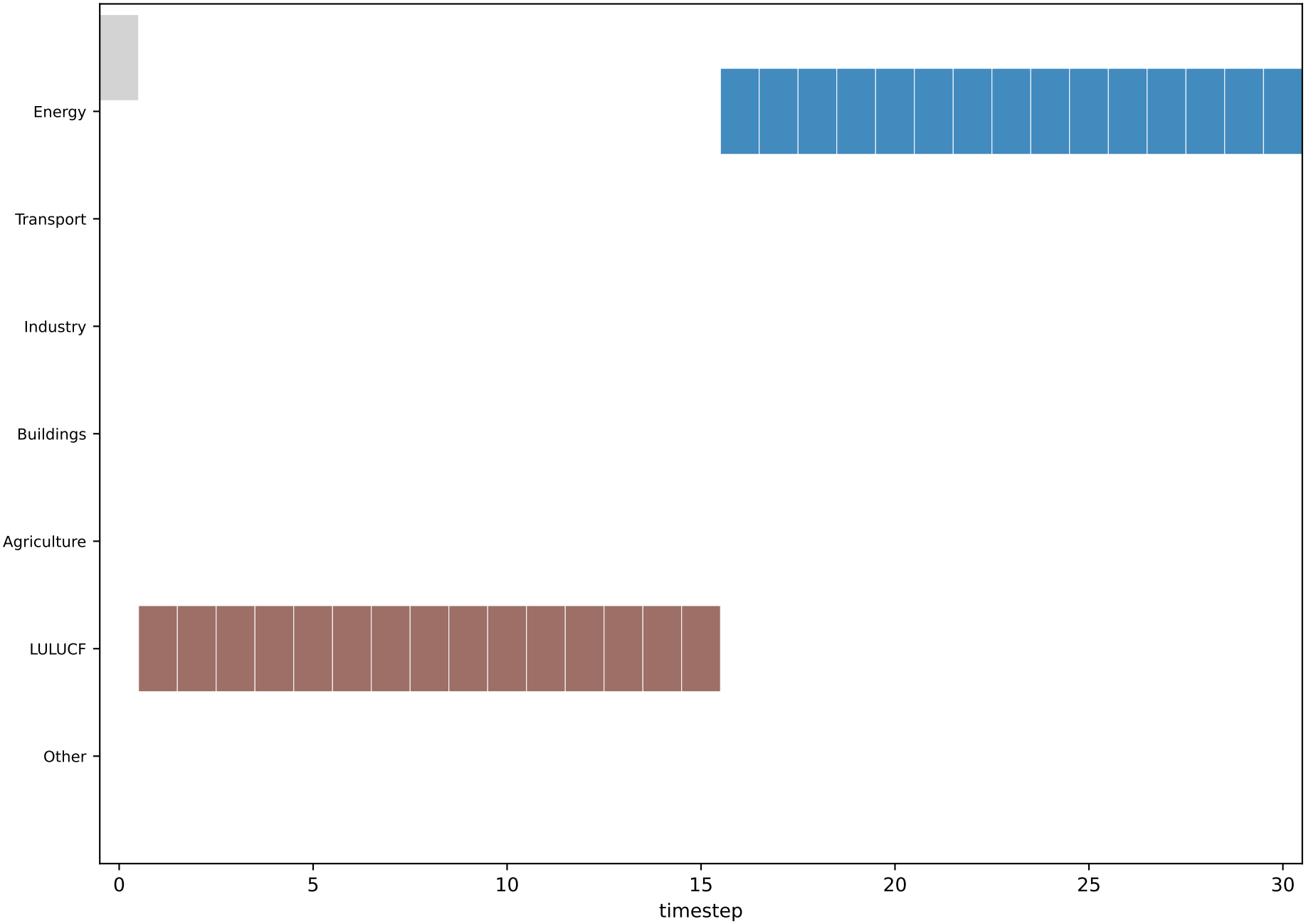
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 21 | Helmsman σ lineage timeline

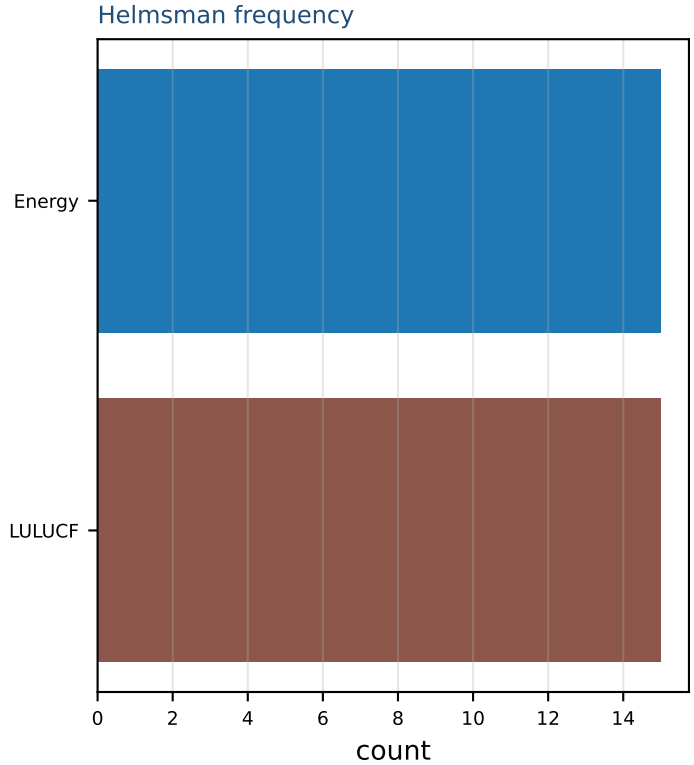
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

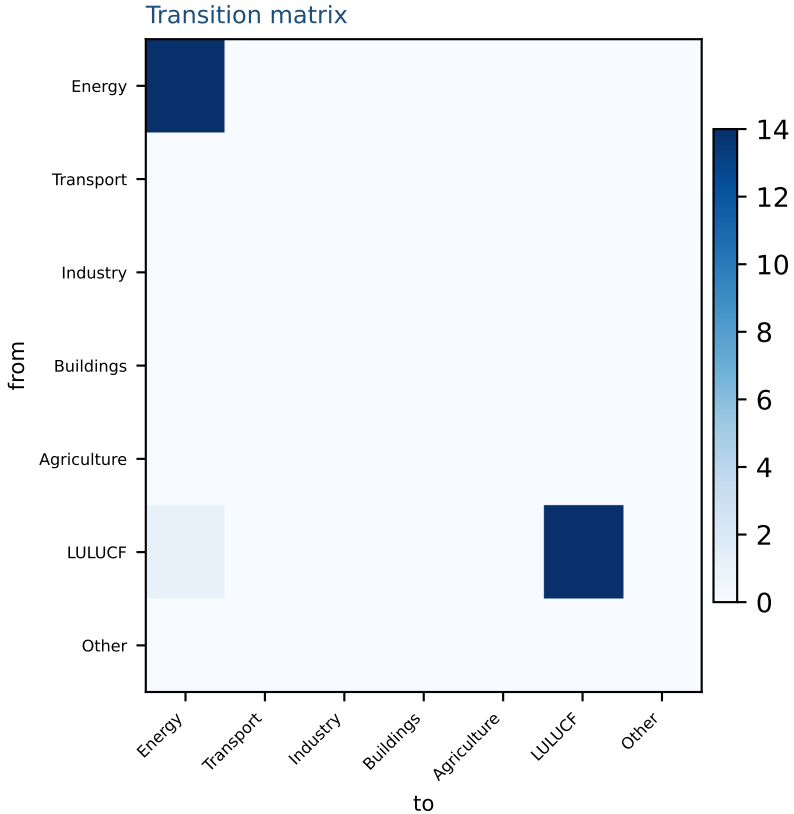


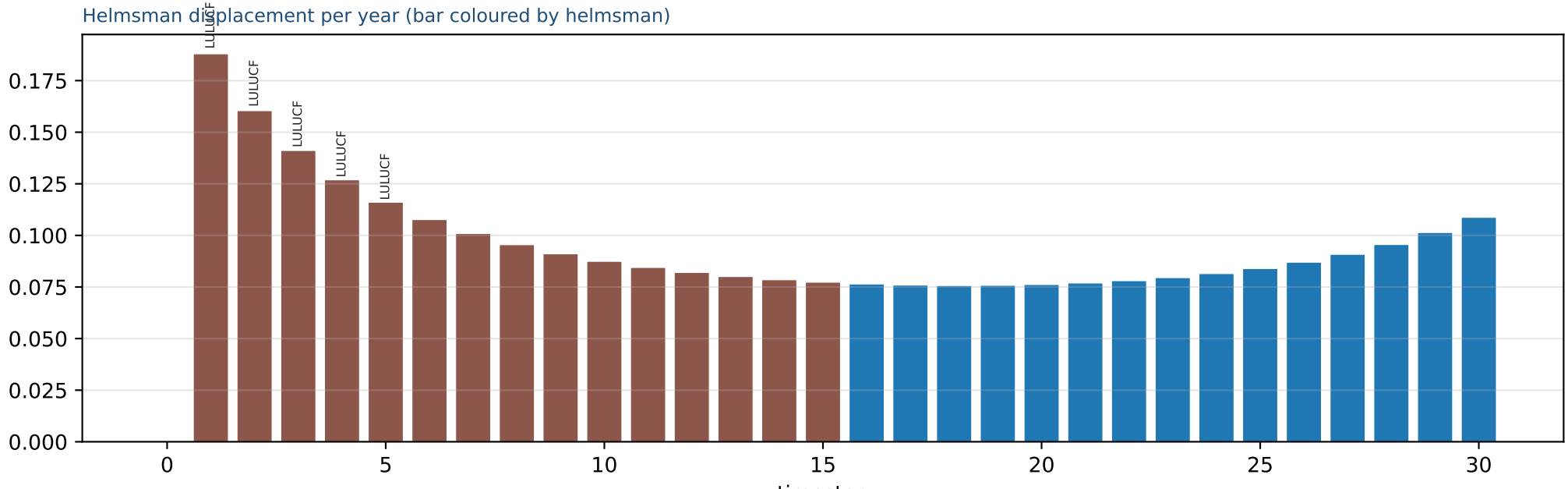
Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 22 | Helmsman σ — frequency · transitions · per-year displacement

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



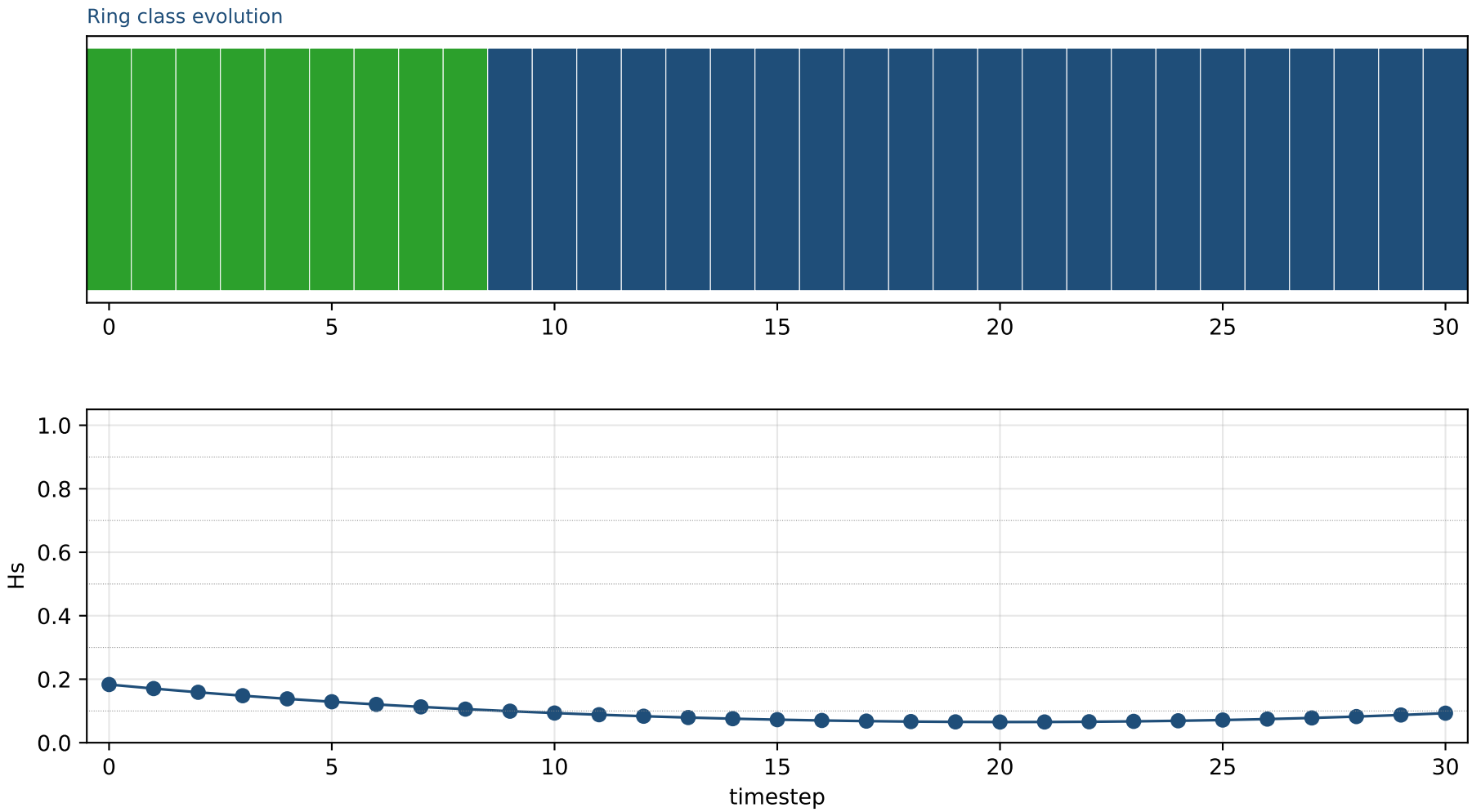




Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 23 | Ring class evolution + Hs

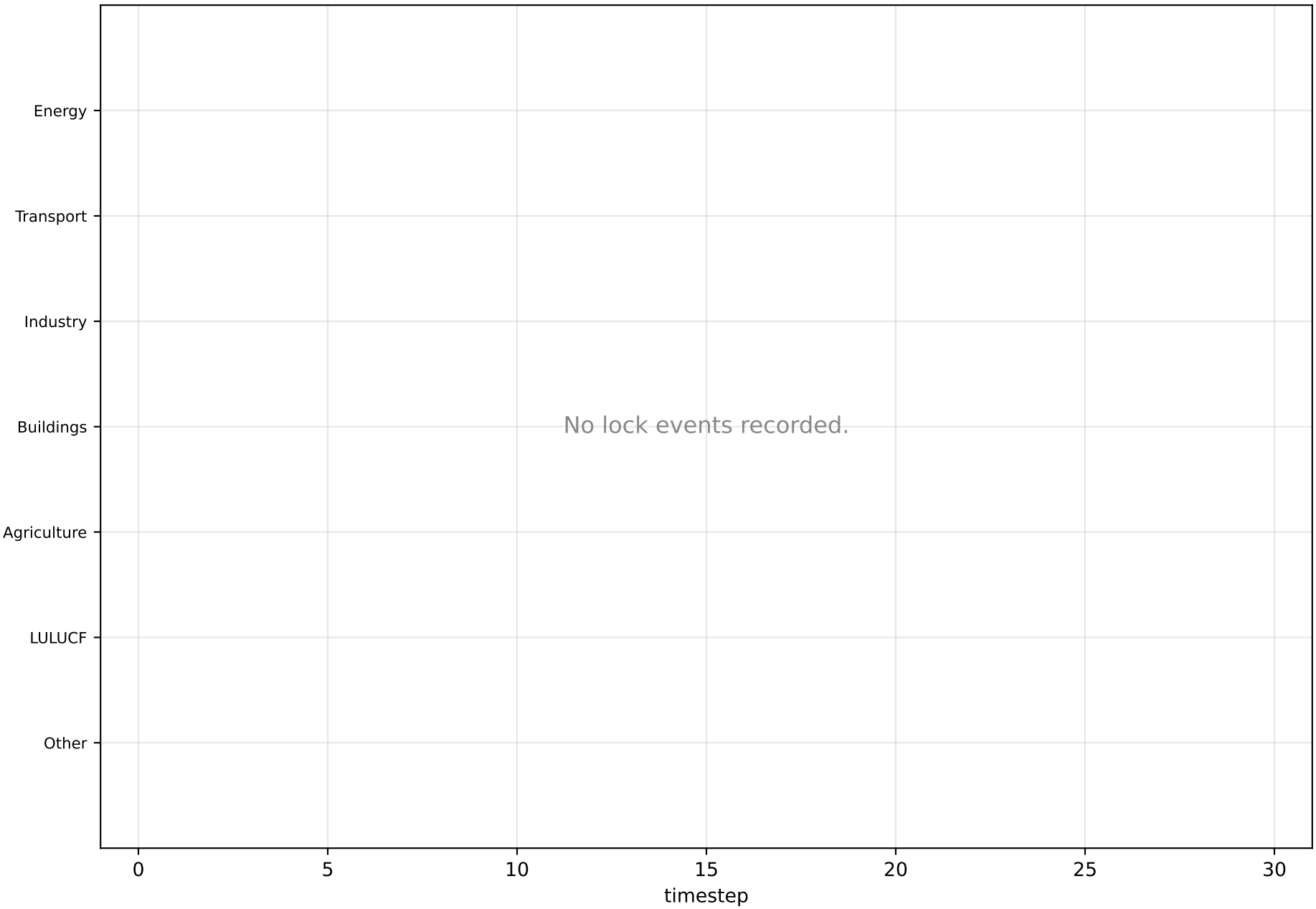
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 24 | Lock event timeline (carriers × timesteps)

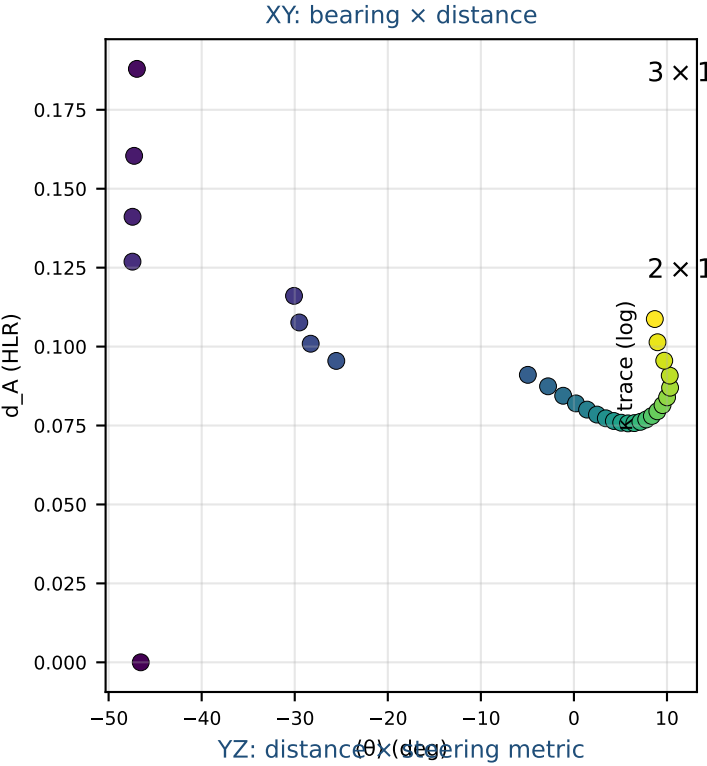
Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

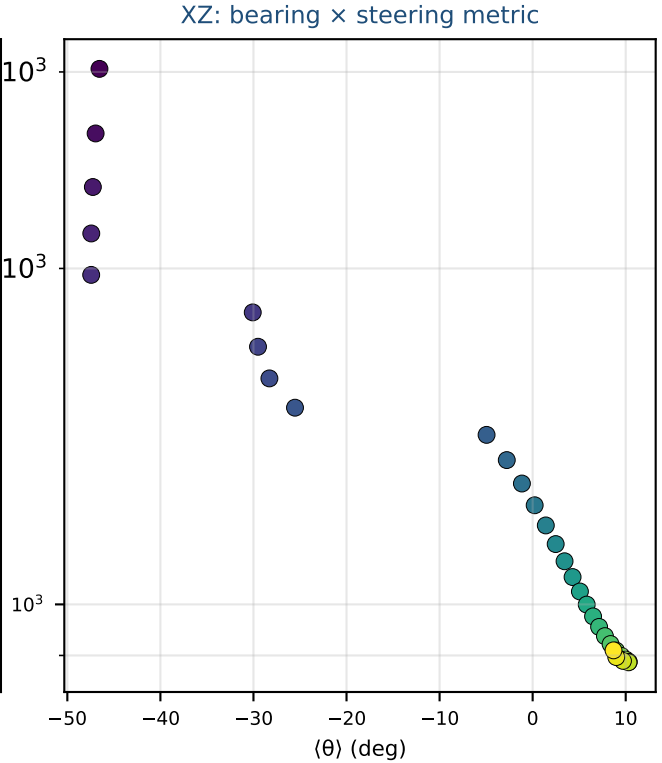


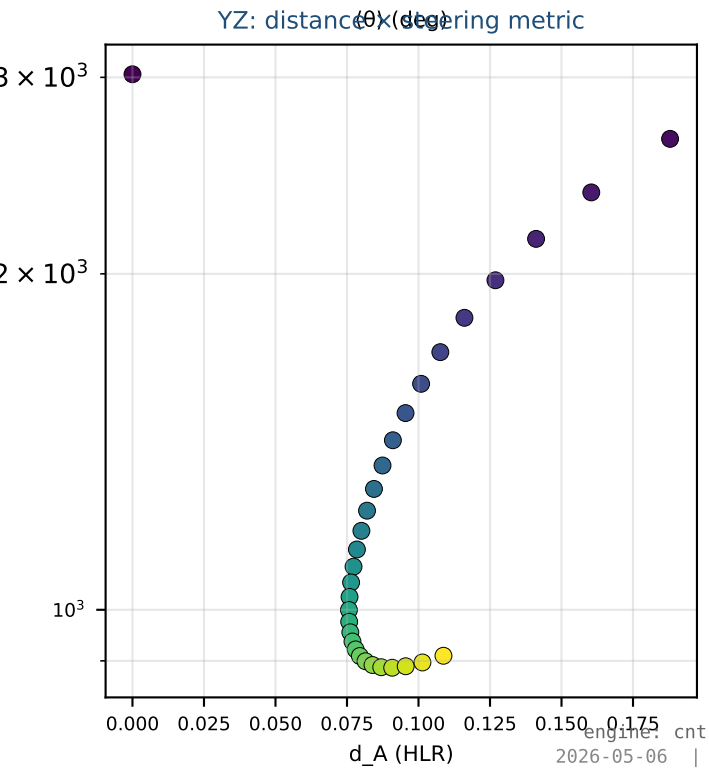
Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 25 | Compositional Bearing Scope (CBS) — 3 orthogonal faces (Math Handbook Ch 16)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2







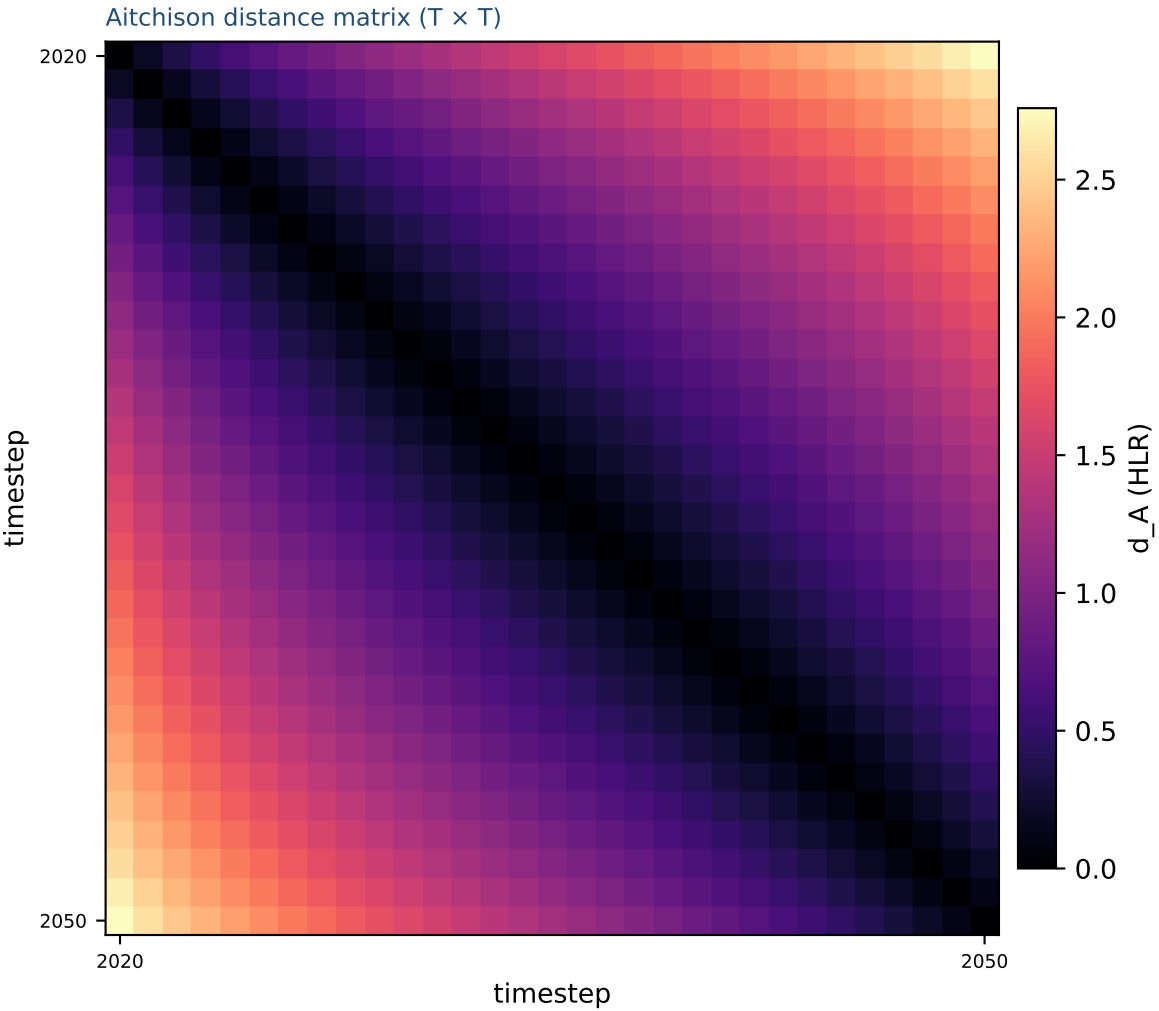
CBS info panel (Math Handbook Ch 16)

```
⟨θ⟩ range  : [-47.4, 10.3]°
d_A range  : [0.0000, 0.1880] HLR
κ range    : [8.87e+02, 3.02e+03]
Hs range   : [0.0652, 0.1833]
ω max      : 3.55°

Each face: one observation per timestep.
Colour = timestep (early=dark, late=bright).
Three faces locate the system in the
(bearing, distance, steering) measurement cube.
```

Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 26 | Pairwise divergence — distance matrix + top-15 ranking

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

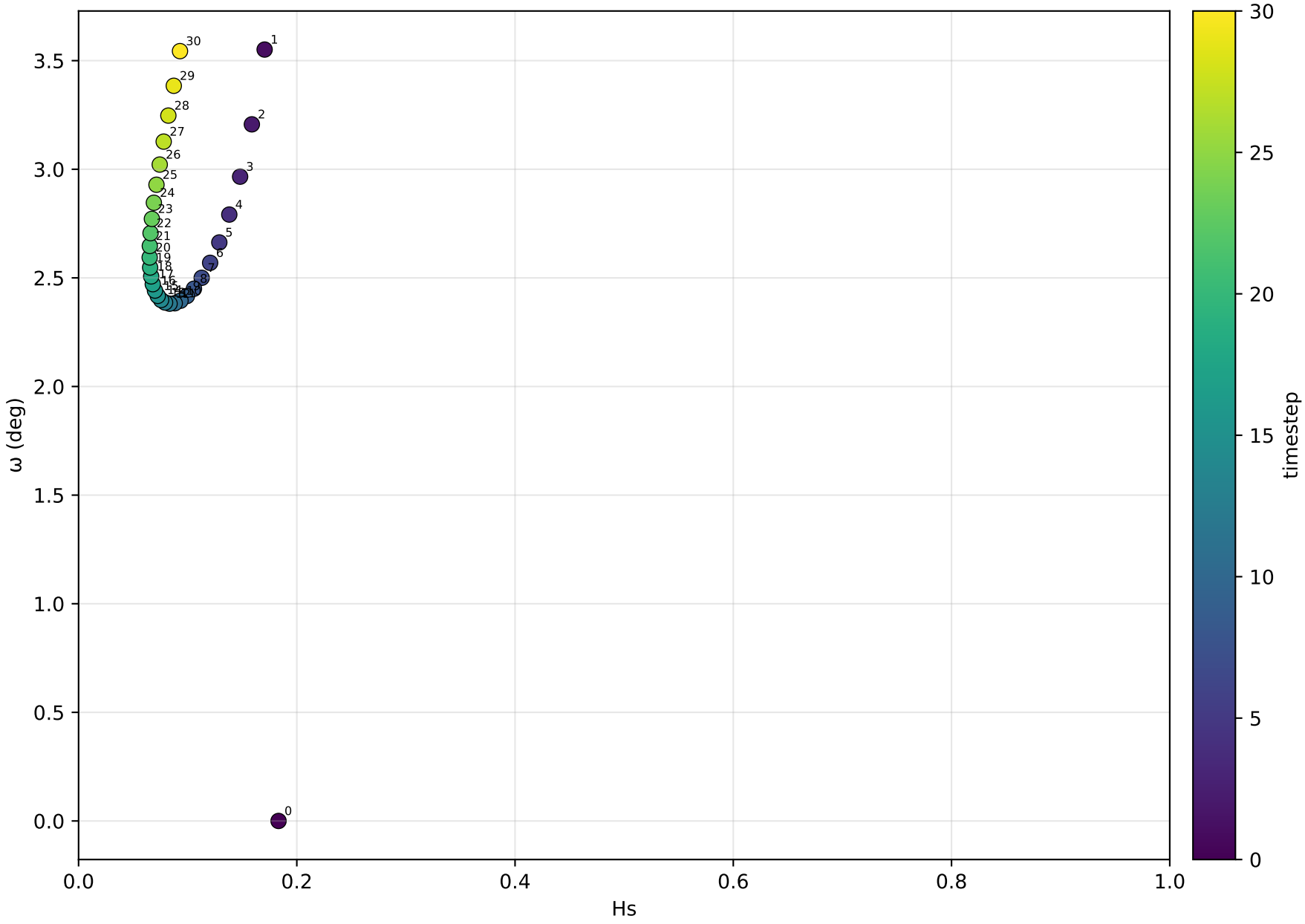


rank	A	B	d_A
1	2020	2050	2.7594
2	2020	2049	2.6636
3	2021	2050	2.5913
4	2020	2048	2.5735
5	2021	2049	2.4941
6	2020	2047	2.4880
7	2022	2050	2.4463
8	2020	2046	2.4060
9	2021	2048	2.4028
10	2022	2049	2.3480
11	2020	2045	2.3269
12	2023	2050	2.3176
13	2021	2047	2.3161
14	2022	2048	2.2557
15	2020	2044	2.2501

Compositional system geometry and dynamics for, IIASA NGFS Phase-4 NZ-2050 sectors 2020-2050

Plate 27 | ω vs Hs scatter (XY test)

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



```
T = 31,  D = 7
carriers: Energy, Transport, Industry, Buildings, Agriculture, LULUCF, Other

Navigation (System Course Plot):
  net_distance     = 2.7594 HLR
  path_length      = 2.8803 HLR
  course_directness= 0.9580

ω (deg)        : min 0.00  mean 2.65  max 3.55
Hs             : min 0.0652  mean 0.0951  max 0.1833
Aitchison norm : min 1.6399  mean 1.8549  max 2.5114
κ trace        : min 887.33  max 3019.09
condition #    : min 5.33e+00  max 2.10e+01

Bearings: range [-179.06, 179.03]°,  std 79.88°
Lock events    : 0
Pairs          : 21
|r| > 0.7 pairs: 21

Helmsman frequency (top 5):
  LULUCF          15  (48.4%)
  Energy          15  (48.4%)
  (none)           1  (3.2%)

Variation matrix trace : 0.0000
```